

$(\Sigma^3, [g])$: conformal 3-mfd

$g \in [g]$, $\mathbb{C}$ -bilinear に拡張

$$\mathbb{C}T^*\Sigma \setminus \{0\} \supset \hat{M}^7 := \{ \xi = \xi_i dx^i \mid g(\xi, \xi) = 0, \xi \neq 0 \}$$

(cpx null covectors)

$$\mathbb{P}(\mathbb{C}T^*\Sigma) \supset M^5 := \hat{M}/\mathbb{C}^* = \{ [\xi] \mid g(\xi, \xi) = 0 \}$$

$$\begin{array}{c} \downarrow \hat{\pi} \\ \downarrow \pi \\ \Sigma \end{array}$$

$$\text{fiber: } C_{x_0} := \pi^{-1}(x_0) \cong \left\{ \left[\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} \right] \in \mathbb{P}^2 \mid \xi_1^2 + \xi_2^2 + \xi_3^2 = 0 \right\}$$

$$\cong \mathbb{P}^1 \left(\ni \left[\begin{pmatrix} \xi \\ \eta \end{pmatrix} \right] \mapsto \begin{bmatrix} -2i\xi\eta \\ \xi^2 + \eta^2 \\ i(\xi^2 - \eta^2) \end{bmatrix} \right)$$

$\hat{c} := (\xi_i dx^i)|_{\mathbb{C}T\hat{M}}$: tautological 1-form on $\hat{M}$

$$T^{1,0}M (= \overline{T^{1,0}M}) := \hat{\pi}_* \{ v \in \mathbb{C}T\hat{M} \mid v \lrcorner d\hat{c} = 0 \} \subset \mathbb{C}T\hat{M}$$

$\mathbb{C}^*-inv$ とする

Prop & Def (LeBrun 1984) $\rightarrow$ Levi form の 7.7.2 (1.1)

$(M^5, T^{1,0}M)$ は Lorentz CR mfd になる

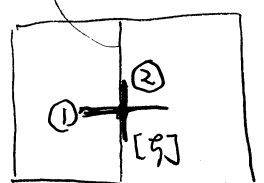
$\supset M^5(\Sigma, [g])$ の twistor CR mfd とし (Σ 上の $\mathbb{P}^1$ -bdl)

$g \in [g]$: fix (x^1, x^2, x^3) : normal coord. centered at $x_0 \in \Sigma$

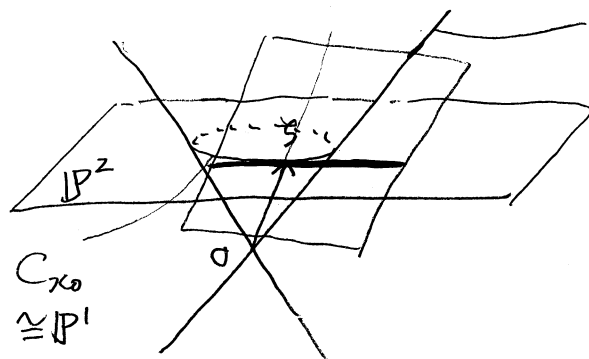
とすると, $[\xi] \in C_{x_0} = \pi^{-1}(x_0)$ に 7.7.1.

$$T^{1,0}_{[\xi]}M = \underbrace{\mathbb{C} \bar{\xi}^i \frac{\partial}{\partial x^i}}_{(1)} \oplus \underbrace{\left(\left\{ w_i \frac{\partial}{\partial \xi_i} \mid \bar{\xi}_i w^i = 0 \right\} / \mathbb{C} \xi_i \frac{\partial}{\partial \xi_i} \right)}_{(2)}$$

C_{x_0}



$$\begin{array}{c} M \\ \downarrow \pi \\ \Sigma \end{array}$$



$$\{ \xi_1^2 + \xi_2^2 + \xi_3^2 = 0 \}$$

$$\mathbb{C}\mathbb{P}^3 \cong \mathbb{C}T_{x_0}^4 \Sigma$$

example (flat model)

(2)

$$\Sigma = S^3 = SU(2) = \left\{ \begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix} \mid |z|^2 + |w|^2 = 1 \right\} \cong \mathbb{R}^3.$$

$$T^*S^3 = SU(2) \times \mathfrak{su}(2), \quad \mathbb{C}T^*S^3 = SU(2) \times \mathfrak{sl}(2, \mathbb{C}).$$

$$X = \begin{pmatrix} i\zeta_1 & -\zeta_2 + i\zeta_3 \\ \zeta_2 + i\zeta_3 & -i\zeta_1 \end{pmatrix} \quad \begin{array}{l} \zeta \in \mathbb{R}^3 \iff X \in \mathfrak{su}(2) \\ \zeta \in \mathbb{C}^3 \iff X \in \mathfrak{sl}(2, \mathbb{C}) \end{array}$$

$$\det X = \zeta_1^2 + \zeta_2^2 + \zeta_3^2.$$

twistor CR mfd:

$$\begin{aligned} M &\cong SU(2) \times \mathbb{P}(\{X \in \mathfrak{sl}(2, \mathbb{C}) \setminus \{0\} \mid \det X = 0\}) \\ &= SU(2) \times \mathbb{P}\left(\left\{ \begin{pmatrix} \xi \\ \eta \end{pmatrix} \begin{pmatrix} \eta & -\xi \end{pmatrix} \mid \begin{pmatrix} \xi \\ \eta \end{pmatrix} \neq 0 \right\}\right) \\ &\cong SU(2) \times \mathbb{P}^1 \end{aligned}$$

$$F: M \cong \underbrace{SU(2)}_U \times \underbrace{\mathbb{P}^1}_W \hookrightarrow \mathbb{P}^3$$

$$(A, [\begin{pmatrix} \xi \\ \eta \end{pmatrix}]) \mapsto \left[\begin{pmatrix} w_0 \\ w_1 \end{pmatrix}, \begin{pmatrix} w_2 \\ w_3 \end{pmatrix} \right] := \left[A \begin{pmatrix} \xi \\ \eta \end{pmatrix}, \begin{pmatrix} \xi \\ \eta \end{pmatrix} \right]$$

$$1 \cong \mathbb{R}^4, \quad M \stackrel{\mathbb{C}\mathbb{R}}{\cong} \left\{ [(w_0, w_1, w_2, w_3)] \in \mathbb{P}^3 \mid |w_0|^2 + |w_1|^2 - |w_2|^2 - |w_3|^2 = 0 \right\}$$

$\cong \mathbb{R}^3$

@ Non-embeddability thm

Thm (LeBrun 1984)

$(\Sigma^3, [g])$: conf 3-mfd $(M^5, T^{1,0}M)$: twistor CR mfd

M is a 3-mfd T is CR embeddable $\iff$

$(\Sigma, [g])$ is C^ω conf mfd $\iff$ 共形同値 $\iff$ 存在.

③

Lorentz CR 5-mfd 動作する:

$$V := \{x \in \Sigma \mid [g] \text{ is } x \text{ on nbd } \Sigma'' \text{ conformally flat}\}$$

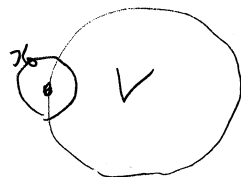
$$\Leftrightarrow \text{Cotton tensor} \equiv 0$$

$$V \neq \emptyset, \Sigma \text{ is a } \mathbb{Z}[g]\text{-module } (\Sigma, [g]) \text{ is free. } x_0 \in \partial V \text{ is not a c.p.}$$

[9] $C^{\omega} = \mathbb{R}^3 \mathbb{R}^n$.

$$\Leftarrow \text{or } \exists p \in \pi^{-1}(x_0) \text{ s.t. } M|_p \text{ is nbd } Z'$$

CR non-embeddable



② CR contact form τ on M

$$M \perp \text{ or CR line ball } L \rightarrow M \perp$$

$$L^{-1} := \hat{M} \times_{\mathbb{Q}^*} \mathbb{C} \quad \text{により定義} \quad (\text{where } \hat{M} = L^{-1} \{303\})$$

$x \in \Sigma$ a fiber $C_x = \pi^{-1}(x) \cong \mathbb{P}^1$ is a limit of Σ .

$$L|_{C_X} \cong \mathcal{O}_{P(C_{T_X}^* \Sigma)}(1)|_{C_X} \cong \mathcal{O}_{C_X}(2).$$

$$\hat{c} \left(\zeta_i \frac{\partial}{\partial \zeta_i} \right) = 0$$

$$\frac{\hat{\cdot}}{L} : \text{homog of deg } 1$$

f). ML of L-valued CR 1-form

$$\tau \in \Gamma(\Lambda_{M^0}^{1,0} \otimes L), \quad \bar{\partial}_b \tau = 0$$

が定まり, $\tau \wedge d\tau \in \Gamma(K_M \otimes L^{\otimes 2})$

17 non-vanishing $(\tau : \text{CR contact form})$

$$(\sim L \approx K_M^{-\frac{1}{2}})$$

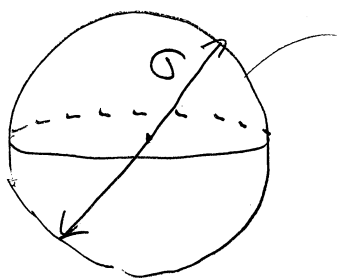
© CR anti-involution σ on M

$$\sigma: M \rightarrow M$$

$$\sigma^2 = \text{id}, \quad \sigma_*(T^{1,0}M) = T^{0,1}M$$

$$[\zeta_i dx^i] \mapsto [\bar{\zeta}_i dx^i]$$

$$\mathcal{T}(C_n) = C_n$$


$$C_n \cong \mathbb{P}^1 \text{ 上 } \mathbb{Z}^n \text{ 付}$$

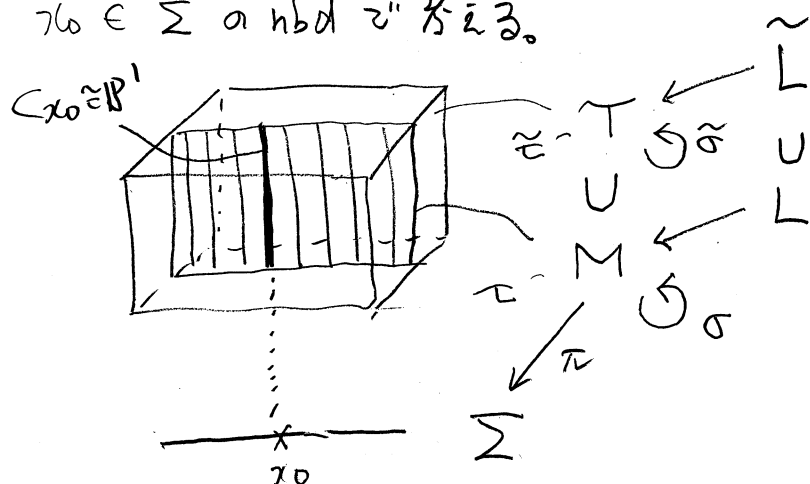
antipodal map

$$\begin{bmatrix} \left(\frac{5}{7} \right) \end{bmatrix} \xrightarrow{\sigma} \begin{bmatrix} \left(\frac{7}{5} \right) \end{bmatrix}$$

• sketch of pf of thm

(4)

$x_0 \in \Sigma$ a hbd τ 存在.



L, τ, σ は

- $\tilde{L} \rightarrow T$: holo line bdl
 $\begin{pmatrix} S_{11} \\ K_T^{-\frac{1}{2}} \end{pmatrix}$
- $\tilde{\tau} \in \Omega^1(\tilde{L})$: holo contact form on T
- $\tilde{\sigma} : T \rightarrow T$: anti-involution

に拡張される

$C_{x_0} \subset T$ の normal bdl $N_{C_{x_0}}$ は M 内で 定義される.

$$N_{C_{x_0}} \cong \mathcal{O}(1) \oplus \mathcal{O}(1)$$

とされる (C_{x_0} : twistor line)

Kodaira $\rightarrow \exists X^4$: 4-dim family of deformation of C_{x_0} in T
 (Cpx spacetime)

$$X \ni x \longleftrightarrow C_x (\cong \mathbb{P}^1) \subset T$$

$$(T_{x,1}^{1,0} X \cong H^0(C_x, N_{C_x}) \cong \mathbb{C}^4)$$

$$\iota : \Sigma \hookrightarrow X \quad : \text{embedding.}$$

$$\downarrow \quad \downarrow$$

$$x \mapsto C_x = \pi^{-1}(x)$$

$Y^3 := \{x \in X \mid \tilde{\tau}|_{T^{1,0}C_x} = 0\}$ は X の hypersurface である.

$$T_x^{1,0} Y \subset T_x^{1,0} X \cong H^0(C_x, N_{C_x}) \xrightarrow{\tilde{\tau}} H^0(C_x, \tilde{L})$$

より $T_x^{1,0} Y \cong H^0(C_x, \tilde{L}) \cong H^0(C_x, \mathcal{O}(2))$ である.

$x \in \Sigma$ のとき $\tilde{\tau}|_{T^{1,0}C_x} = \tau|_{T^{1,0}C_x} = 0$ より $(\iota(\Sigma)) \subset Y$.

Y 上の holo conf str $[g_Y] \in$

$v \in T_x^{\perp 0} Y \cong H^0(C_x, \mathcal{O}(2))$ が "null"

$\Leftrightarrow v$ は C_x 上の 2 重解 Σ かつ (判別式 = 0)

により定まる

anti-involution $\sigma: Y \rightarrow Y$ $C_{\sigma(x)} := \sigma(C_x)$

に関する real form Σ

$$Y^\sigma := \{ x \in Y \mid \sigma(x) = x \}$$

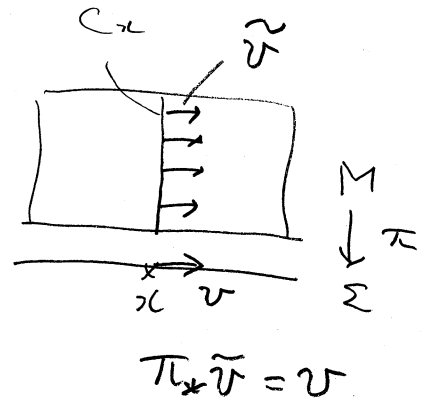
とすると, $\underbrace{L(\Sigma)}_{\text{実 3 次元}} \subset Y^\sigma$ より $L(\Sigma) = Y^\sigma$ (Y は Σ の複素化)

$\mathbb{C}TY^\sigma \cong (T^{\perp 0}Y)|_{Y^\sigma} \ni \uparrow \subset \mathbb{C}TY^\sigma = \text{null cone}$
が定まり, $(Y^\sigma, [g_{Y^\sigma}]) : \mathbb{C}^\omega \text{ conf mfd}$ となる.

$v \in \mathbb{C}T_x \Sigma$ に対して

$L_* v \in \mathbb{C}T_{L(x)} Y^\sigma$ が "null"

$\Leftrightarrow \tau(\tilde{v})$ が C_x 上の 2 重解 Σ かつ



normal coord $\Leftrightarrow v^i \xi_i$ が $\mathbb{P}^1$ 上の 2 重解 Σ かつ

$$\begin{aligned} & v^1(-2i\xi\eta) + v^2(\xi^2 + \eta^4) + v^3 \cdot i(\xi^2 - \eta^4) \\ &= (v^2 + iv^3)\xi^2 - 2iv^1\xi\eta + (v^2 - iv^3)\eta^2 \end{aligned}$$

$$\Leftrightarrow (v^1)^2 + (v^2)^2 + (v^3)^2 = 0$$

$$\Leftrightarrow v : \text{null}$$

より, $L: (\Sigma, [g]) \rightarrow (Y^\sigma, [g_{Y^\sigma}])$ は conf. equiv //